Research Analysis Report

Research Topic: Multi-Agent System

Total Papers: 5

Generated By: Report Generation Agent

Generated At: 2026-08-25 18:13:59

Executive Summary

This report provides a technical overview of 'Multi-Agent System' including methodologies, research gaps and experiment planning.

Literature Review

Introduction: This literature review summarizes findings from 5 research papers related to the selected topic. It highlights research trends, methodologies, key contributions, future directions, and overall observations.

Paper Summaries

• {'title': 'Magentic-One: A Generalist Multi-Agent System for Solving Complex Tasks', 'year': 2024, 'summary': "Modern AI agents, driven by advances in large foundation models, promise to enhance our productivity and transform our lives by augmenting our knowledge and capabilities. To achieve this vision, AI agents must effectively plan, perform multi-step reasoning and actions, respond to novel observations, and recover from errors, to successfully complete complex tasks across a wide range of scenarios. In this work, we introduce Magentic-One, a high-performing open-source agentic system for solving such tasks. Magentic-One uses a multi-agent architecture where a lead agent, the Orchestrator, plans, tracks progress, and re-plans to recover from errors. Throughout task execution, the Orchestrator directs other specialized agents to perform tasks as needed, such as operating a web browser, navigating local files, or writing and executing Python code. We show that Magentic-One achieves statistically competitive performance to the state-of-the-art on three diverse and challenging agentic benchmarks: GAIA, AssistantBench, and WebArena. Magentic-One achieves these results without modification to core agent capabilities or to how they collaborate, demonstrating progress towards generalist agentic systems. Moreover, Magentic-One's modular design allows agents to be added or removed from the team without additional prompt tuning or training, easing development and making it extensible to future scenarios. We provide an open-source implementation of"}

• {'title': 'PsySafe: A Comprehensive Framework for Psychological-based Attack, Defense, and Evaluation of Multi-agent System Safety', 'year': 2024, 'summary': "Multi-agent systems, when enhanced with Large Language Models (LLMs), exhibit profound capabilities in collective intelligence. However, the potential misuse of this intelligence for malicious purposes presents significant risks. To date, comprehensive research on the safety issues associated with multi-agent systems remains limited. In this paper, we explore these concerns through the innovative lens of agent psychology, revealing that the dark psychological states of agents constitute a significant threat to safety. To tackle these concerns, we propose a comprehensive framework (PsySafe) grounded in agent psychology, focusing on three key areas: firstly, identifying how dark personality traits in agents can lead to risky behaviors; secondly, evaluating the safety of multi-agent systems from the psychological and behavioral perspectives, and thirdly, devising effective strategies to mitigate these risks. Our experiments reveal several intriguing phenomena, such as the collective dangerous behaviors among agents, agents' self-reflection when engaging in dangerous behavior, and the correlation between agents' psychological assessments and dangerous behaviors. We anticipate that our framework and observations will provide valuable insights for further research into the safety of multi-agent systems. We will make our data and code publicly accessible at https://github.com/AI4Good24/PsySafe."}

• {'title': 'Towards smart factory for industry 4.0: a self-organized multi-agent system with big data based feedback and coordination', 'year': 2016, 'summary': ''}

• {'title': 'MIRIX: Multi-Agent Memory System for LLM-Based Agents', 'year': 2025, 'summary': "Although memory capabilities of AI agents are gaining increasing attention, existing solutions remain fundamentally limited. Most rely on flat, narrowly scoped memory components, constraining their ability to personalize, abstract, and reliably recall user-specific information over time. To this end, we introduce MIRIX, a modular, multi-agent memory system that redefines the future of AI memory by solving the field's most critical challenge: enabling language models to truly remember. Unlike prior approaches, MIRIX transcends text to embrace rich visual and multimodal experiences, making memory genuinely useful in real-world scenarios. MIRIX consists of six distinct, carefully structured memory types: Core, Episodic, Semantic, Procedural, Resource Memory, and Knowledge Vault, coupled with a multi-agent framework that dynamically controls and coordinates updates and retrieval. This design enables agents to persist, reason over, and accurately retrieve diverse, long-term user data at scale. We validate MIRIX in two demanding settings. First, on ScreenshotVQA, a challenging multimodal benchmark comprising nearly 20,000 high-resolution computer screenshots per sequence, requiring deep contextual understanding and where no existing memory systems can be applied, MIRIX achieves 35% higher accuracy than the RAG baseline while reducing storage requirements by 99.9%. Second, on LOCOMO, a long-form conversation benchmark with single-modal textual input, MIRIX attains state-of-the-art performance"}

• {'title': 'A multi-agent system for automating scientific discovery', 'year': 2026, 'summary': 'Scientific discovery is driven by the iterative process of observation, hypothesis generation, experimentation and data analysis. Despite recent advancements in applying artificial intelligence (AI) to biology, no system has yet automated all these stages1, 2–3. Here we introduce Robin, a multi-agent system capable of fully automating both hypothesis generation and data analysis for experimental biology. By integrating literature search agents with data analysis agents, Robin can generate hypotheses, propose experiments, interpret experimental results and generate updated hypotheses, achieving a semi-autonomous approach to scientific discovery. By applying this system, we were able to identify promising therapeutic candidates for dry age-related macular degeneration, the major cause of blindness in the developed world4,5. Robin proposed enhancing retinal pigment epithelium phagocytosis as a therapeutic strategy, and identified and confirmed in vitro efficacy for ripasudil and KL001. Ripasudil is a clinically used Rho kinase inhibitor that, to our knowledge, has never previously been proposed for the treatment of dry age-related macular degeneration. To elucidate the mechanism of ripasudil-induced upregulation of phagocytosis, Robin then proposed and analysed a follow-up RNA sequencing experiment, which revealed upregulation of ABCA1, which encodes a lipid efflux pump and represents a possible novel target. All hypotheses, experimental directions, data analyses and'}

Research Trends

• advance

• advancement

• agent

• although

• analysis

• applying

• attention

- augmenting
- capability
- collective
- data
- despite
- discovery
- driven
- enhance
- enhanced
- exhibit
- existing
- experimentation
- flat
- foundation
- fundamentally
- gaining
- generation
- however
- hypothesis
- increasing
- intelligence
- iterative
- knowledge
- language
- large
- limited
- lives
- llms
- memory
- model
- modern
- multi-agent
- narrowly
- observation
- potential
- process
- productivity
- profound

- promise
- recent
- rely
- remain
- scientific
- solution
- system
- transform
- when

Research Gaps

- Moreover, Magentic-One's modular design allows agents to be added or removed from the team without additional prompt tuning or training, easing development and making it extensible to future scenarios
- To this end, we introduce MIRIX, a modular, multi-agent memory system that redefines the future of AI memory by solving the field's most critical challenge: enabling language models to truly remember
- We anticipate that our framework and observations will provide valuable insights for further research into the safety of multi-agent systems

Future Scope

- Moreover, Magentic-One's modular design allows agents to be added or removed from the team without additional prompt tuning or training, easing development and making it extensible to future scenarios
- To this end, we introduce MIRIX, a modular, multi-agent memory system that redefines the future of AI memory by solving the field's most critical challenge: enabling language models to truly remember
- We anticipate that our framework and observations will provide valuable insights for further research into the safety of multi-agent systems

Conclusion: Overall, the reviewed studies demonstrate significant progress in this research area while highlighting several opportunities for future work and further experimentation.

Methodology Comparison

Total Papers: 5

Comparison

- {'title': 'Magentic-One: A Generalist Multi-Agent System for Solving Complex Tasks', 'research_area': 'Agentic AI', 'research_problem': 'Modern AI agents, driven by advances in large foundation models, promise to enhance our productivity and transform our lives by augmenting our knowledge and capabilities', 'methodology': ['Architecture', 'Model', 'System'], 'keywords': ['modern', 'agent', 'driven', 'advance', 'large', 'foundation', 'model', 'promise', 'enhance', 'productivity', 'transform', 'lives', 'augmenting', 'knowledge', 'capability'], 'year': 2024, 'citation_count': 247, 'quality_score': 80, 'quality_classification': 'Very Good'}

- {'title': 'PsySafe: A Comprehensive Framework for Psychological-based Attack, Defense, and Evaluation of Multi-agent System Safety', 'research_area': 'Agentic AI', 'research_problem': 'Multi-agent systems, when enhanced with Large Language Models (LLMs), exhibit profound capabilities in collective intelligence', 'methodology': ['Framework', 'Model', 'System', 'Llm'], 'keywords': ['multi-agent', 'system', 'when', 'enhanced', 'large', 'language', 'model', 'llms', 'exhibit', 'profound', 'capability', 'collective', 'intelligence', 'however', 'potential'], 'year': 2024, 'citation_count': 98, 'quality_score': 70, 'quality_classification': 'Very Good'}
- {'title': 'Towards smart factory for industry 4.0: a self-organized multi-agent system with big data based feedback and coordination', 'research_area': 'Agentic AI', 'research_problem': 'Not Available', 'methodology': ['Not Identified'], 'keywords': [], 'year': 2016, 'citation_count': 1263, 'quality_score': 50, 'quality_classification': 'Good'}
- {'title': 'MIRIX: Multi-Agent Memory System for LLM-Based Agents', 'research_area': 'Agentic AI', 'research_problem': 'Although memory capabilities of AI agents are gaining increasing attention, existing solutions remain fundamentally limited', 'methodology': ['Framework', 'Model', 'System', 'Approach', 'Rag'], 'keywords': ['although', 'memory', 'capability', 'agent', 'gaining', 'increasing', 'attention', 'existing', 'solution', 'remain', 'fundamentally', 'limited', 'rely', 'flat', 'narrowly'], 'year': 2025, 'citation_count': 146, 'quality_score': 90, 'quality_classification': 'Excellent'}
- {'title': 'A multi-agent system for automating scientific discovery', 'research_area': 'Agentic AI', 'research_problem': 'Scientific discovery is driven by the iterative process of observation, hypothesis generation, experimentation and data analysis', 'methodology': ['System', 'Approach'], 'keywords': ['scientific', 'discovery', 'driven', 'iterative', 'process', 'observation', 'hypothesis', 'generation', 'experimentation', 'data', 'analysis', 'despite', 'recent', 'advancement', 'applying'], 'year': 2026, 'citation_count': 98, 'quality_score': 80, 'quality_classification': 'Very Good'}

Highest Cited Paper

title: Towards smart factory for industry 4.0: a self-organized multi-agent system with big data based feedback and coordination

authors: ['Shiyong Wang', 'J. Wan', 'Daqiang Zhang', 'Di Li', 'Chunhua Zhang']

year: 2016

citation_count: 1263

summary:

research_problem: Not Available

methodology: ['Not Identified']

key_contributions: []

future_work: []

keywords: []

research_area: Agentic AI

paper_score: 50

paper_quality: Good

Latest Paper

title: A multi-agent system for automating scientific discovery

authors: ['A. Ghareeb', 'Benjamin Chang', 'L. Mitchener', 'Angela Yiu', 'Caralyn J. Szostkiewicz', 'D. Shved', 'Gavin Gyimesi', 'Jon M. Laurent', 'S. Wright', 'Muhammed Razzak', 'Andrew D. White', 'S. Finnemann', 'Michaela M. Hinks', 'S. Rodriques']

year: 2026

citation_count: 98

summary: Scientific discovery is driven by the iterative process of observation, hypothesis generation, experimentation and data analysis. Despite recent advancements in applying artificial intelligence (AI) to biology, no system has yet automated all these stages1, 2–3. Here we introduce Robin, a multi-agent system capable of fully automating both hypothesis generation and data analysis for experimental biology. By integrating literature search agents with data analysis agents, Robin can generate hypotheses, propose experiments, interpret experimental results and generate updated hypotheses, achieving a semi-autonomous approach to scientific discovery. By applying this system, we were able to identify promising therapeutic candidates for dry age-related macular degeneration, the major cause of blindness in the developed world4,5. Robin proposed enhancing retinal pigment epithelium phagocytosis as a therapeutic strategy, and identified and confirmed in vitro efficacy for ripasudil and KL001. Ripasudil is a clinically used Rho kinase inhibitor that, to our knowledge, has never previously been proposed for the treatment of dry age-related macular degeneration. To elucidate the mechanism of ripasudil-induced upregulation of phagocytosis, Robin then proposed and analysed a follow-up RNA sequencing experiment, which revealed upregulation of ABCA1, which encodes a lipid efflux pump and represents a possible novel target. All hypotheses, experimental directions, data analyses and

research_problem: Scientific discovery is driven by the iterative process of observation, hypothesis generation, experimentation and data analysis

methodology: ['System', 'Approach']

key_contributions: ['Here we introduce Robin, a multi-agent system capable of fully automating both hypothesis generation and data analysis for experimental biology', 'By integrating literature search agents with data analysis agents, Robin can generate hypotheses, propose experiments, interpret experimental results and generate updated hypotheses, achieving a semi-autonomous approach to scientific discovery', 'By applying this system, we were able to identify promising therapeutic candidates for dry age-related macular degeneration, the major cause of blindness in the developed world4,5']

future_work: []

keywords: ['scientific', 'discovery', 'driven', 'iterative', 'process', 'observation', 'hypothesis', 'generation', 'experimentation', 'data', 'analysis', 'despite', 'recent', 'advancement', 'applying']

research_area: Agentic AI

paper_score: 80

paper_quality: Very Good

Common Methodologies

• Approach

• Architecture

• Framework

• Llm

• Model

• Not Identified

• Rag

• System

Research Areas

• Agentic AI

Common Keywords

- advance
- advancement
- agent
- although
- analysis
- applying
- attention
- augmenting
- capability
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Methodology Frequency

System: 4

Model: 3

Framework: 2

Approach: 2

Architecture: 1

Llm: 1

Not Identified: 1

Rag: 1

Most Common Methodology: System

Quality Distribution

Very Good: 3

Good: 1

Excellent: 1

Publication Years

2016: 1

2024: 2

2025: 1

2026: 1

Citation Statistics

highest: 1263

lowest: 98

average: 370.4

total: 1852

Comparison Summary

total_papers: 5

most_common_methodology: System

research_areas: ['Agentic AI']

quality_distribution: {'Very Good': 3, 'Good': 1, 'Excellent': 1}

citation_statistics: {'highest': 1263, 'lowest': 98, 'average': 370.4, 'total': 1852}

highest_cited_title: Towards smart factory for industry 4.0: a self-organized multi-agent system with big data based feedback and coordination

latest_paper_title: A multi-agent system for automating scientific discovery

Research Gap Analysis

Total Papers: 5

Research Areas

• Agentic AI

Common Keywords

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Future Work

• Moreover, Magentic-One's modular design allows agents to be added or removed from the team without additional prompt tuning or training, easing development and making it extensible to future scenarios

• To this end, we introduce MIRIX, a modular, multi-agent memory system that redefines the future of AI memory by solving the field's most critical challenge: enabling language models to truly remember

• We anticipate that our framework and observations will provide valuable insights for further research into the safety of multi-agent systems

Research Area Distribution

Agentic AI: 5

Keyword Frequency

capability: 3

agent: 2

driven: 2

large: 2

model: 2

modern: 1

advance: 1

foundation: 1

promise: 1

enhance: 1

productivity: 1

transform: 1

lives: 1

augmenting: 1

knowledge: 1

multi-agent: 1

system: 1

when: 1

enhanced: 1

language: 1

llms: 1

exhibit: 1

profound: 1

collective: 1

intelligence: 1

however: 1

potential: 1

although: 1

memory: 1

gaining: 1

increasing: 1

attention: 1

existing: 1

solution: 1

remain: 1

fundamentally: 1

limited: 1

rely: 1

flat: 1

narrowly: 1

scientific: 1

discovery: 1

iterative: 1

process: 1

observation: 1

hypothesis: 1

generation: 1

experimentation: 1

data: 1

analysis: 1

despite: 1

recent: 1

advancement: 1

applying: 1

Research Trends

- capability
- agent
- driven
- large
- model
- modern
- advance

- foundation
- promise
- enhance
- productivity
- transform
- lives
- augmenting
- knowledge

Gap Categories

datasets: []

models: ["To this end, we introduce MIRIX, a modular, multi-agent memory system that redefines the future of AI memory by solving the field's most critical challenge: enabling language models to truly remember"]

evaluation: []

applications: []

general: ["Moreover, Magentic-One's modular design allows agents to be added or removed from the team without additional prompt tuning or training, easing development and making it extensible to future scenarios", 'We anticipate that our framework and observations will provide valuable insights for further research into the safety of multi-agent systems']

Emerging Topics

- capability
- agent
- driven
- large
- model

Recommendations

- Investigate unexplored research directions.

Summary

dominant_area: Agentic AI

top_trend: capability

future_work_items: 3

Experiment Plan

Research Areas

- Agentic AI

Recommended Datasets

• AgentBench

• GAIA Benchmark

• HotpotQA

• Hugging Face Datasets

• OpenAI Evals

Baseline Models

None

Evaluation Metrics

• Accuracy

• Precision

• Recall

• F1-Score

• ROC-AUC

Hardware Requirements

cpu: Intel Core i7 / AMD Ryzen 7

ram: 16 GB

gpu: NVIDIA RTX 3060 or better

storage: 100 GB SSD

Validation Strategy

• Train-Test Split

• K-Fold Cross Validation

• Hyperparameter Tuning

• Statistical Significance Testing

Experimental Workflow

• Collect Dataset

• Preprocess Data

• Train Baseline Models

• Train Proposed Model

• Evaluate Performance

• Compare Results

• Analyze Errors

• Draw Conclusions

Report Summary

Dominant Research Area: Agentic AI

Top Research Trend: capability

Highest Cited Paper: Towards smart factory for industry 4.0: a self-organized multi-agent system with big data based feedback and coordination

Latest Paper: A multi-agent system for automating scientific discovery

Recommended Datasets

- AgentBench
- GAIA Benchmark
- HotpotQA
- Hugging Face Datasets
- OpenAI Evals

Baseline Models

None

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Conclusion

The report has been generated using a personalized multi-agent pipeline. The explanation style was adapted according to the Intermediate profile to improve readability and usefulness.